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Vietnam National University, Hanoi

# INT3405E - Machine Learning

## Lecture 0: Course Introduction

**Hanoi, 09/2026**

# Course Introduction

- **Course Name/Code**: Machine Learning / INT3405
- **#Credits**: 03
- **Lecture Organization:** 2+1 ~ 2-hour lecture + 1-hour Exercise/Practice
- **Course Plan**: 15 weeks ~ 12 lessons + 1 midterm + 2 project presentations
- **Learning outcomes:**
  - Understand basic concepts of machine learning
  - Understand machine learning techniques and algorithms
  - Can use tools/libraries to implement machine learning models
  - Can apply machine learning models in revealing new knowledge
  - Can apply the obtained knowledge to solve real world problems

# Teaching Team

Instructors: Tạ Việt Cường, Phd,
Teaching Assistant

About myself:

Backgrounds: Computer Science

Years in ML: Since 2009

Research Direction: Machine Learning Theory and Applications, Reinforcement Learning, Deep Graph Learning.

# Course Materials

- **Primary Textbook:**
  - Học máy thống Kê, Trần Quốc Long, Tạ Việt Cường, NXBĐHQGHN 2025.
  - Kevin P. Murphy (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press
  - Trevor Hastie , Robert Tibshirani , Jerome Friedman (2001). *The Elements of Statistical Learning, Springer*

- Reference Textbook:
  - Christopher M. Bishop (2006). *Pattern Recognition and Machine Learning, Springer*
  - Ian Goodfellow and Yoshua Bengio and Aaron Courville (2016). *Deep Learning, MIT Press*
  - Athem Alpaydin (2010). *Introduction to Machine Learning, The MIT Press.*

# Basic Knowledge and Prerequisites

- Mathematics Foundations:
  - *Linear Algebra: Understanding vectors, matrices, matrix operations*
  - *Calculus: derivatives, integrals, and partial derivatives*
  - *Probability & Statistics*
  - *Optimization*
- Programming Skills:
  - *Python*
  - *Data Structures and Algorithms*
  - *Software Engineering*

# Course Outline

- Week 1: Introduction
- Week 2: General Concepts
- Week 3: Linear Regression
- Week 4: Classification (1) - Logistic Regression, Bayes
- Week 5: Classification (2) - Decision Tree, KNN
- Week 6: Classification (3) - SVM & Variants
- Week 7: Feature Selection & Model Optimization
- Week 8: Unsupervised Learning
- Week 9: Mid-term Exam
- Week 10: Deep learning (1) - Perceptron, MLP
- Lecture 11: Deep learning (2) - CNN, RNN
- Lecture 12: Introduction to MLOps
- Lecture 13: Reinforcement Learning
- Lecture 14-15: Project Presentation

# Course Assessment

- Class Attendance: 10% (3rd week onwards)
- Mid-term Exam: 30%
  - 10 MCQs + 2-3 practical questions
- Final Project: 60%
  - Group project: A team of 3 students
  - Goal:
    - Choose a problem of a given list, i.e., datasets
    - Implement ML algorithms
    - Evaluate and improve the algorithms
    - Presentation report



# Q&A